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A LOWER ORDOVICIAN GEOMAGNETIC POLE FROM PENNSYLVANIA

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Basalt, diabase, and arkose were sampled from the Middle Ordovician Martinsburg Formation in Lebanon County, Pennsylvania, for paleomagnetic study. The sampled units, situated within blocks emplaced in the Middle Ordovician sedimentary basin during the Taconic orogeny, are believed to be Lower Ordovician. In the course of orogenic deformation, the sampled units and host rock were extensively folded, a factor which had to be accounted for in the derivation of the paleomagnetic pole. The Lower Ordovician virtual geomagnetic pole, calculated after alternating field and thermal cleaning of 26 samples from four sites, lies at 159.8°E longitude and 35.8°N latitude. The agreement of this pole with other Lower Paleozoic poles for the North American plate demonstrates that intense regional deformation need not preclude the recovery of a valid paleomagnetic pole. There does exist a possibility of uncorrected rotation of the block in which the rocks were emplaced.

1. Introduction

The basalt and diabase within the Martinsburg Formation in Lebanon County, Pennsylvania, are among a series of Lower Ordovician blocks within the Middle Ordovician shale [1]. The procedure used to obtain a geomagnetic pole from the igneous rocks is summarized. The nature of the rocks called for careful treatment, particularly concerning the unusually extensive structural corrections.

2. Field and laboratory procedure

The field study was confined to the southern half of the Fredericksburg, Pennsylvania, 7.5′ quadrangle. Basalt, diabase, and a green arkose in contact with diabase were sampled. The samples of basalt proved to be paleomagnetically unstable and were not used in the final analysis of results. Only outcrops which exhibited no apparent non-tectonic displacements were sampled. Due to the extreme deformation which these rocks underwent, structural parameters related to the tectonic position of the rocks were noted. This information, along with data from the map of Bricker [2], was used to estimate the original, undeformed, attitudes of the samples units and host rock and to correct the paleomagnetic data for the sequence of post-intrusive deformations.

The oriented samples were cored and sliced into specimens with diameters of 24.9 mm and lengths of 23.6 mm. The natural remanent magnetization (NRM) of each specimen was estimated by subjecting a specimen to twenty sequences of spins. The standard deviation for the intensity of magnetization was $\pm 6.34 \times 10^{-7}$ emu, not significant in most cases since the range of sample intensities was from 3.75×10^{-5} to 1.92×10^{-2} emu. The standard deviation for the direction of the total magnetization vector was found to be ±0.36°.

In order to determine which method of cleaning would be the most appropriate, several specimens were selected as test samples. Alternating field cleaning was done in a Schonstedt GSD-1 Geophysical Specimen Demagnetizer in steps in 100 Oe up to a peak applied field of 700 Oe. Thermal demagnetization was done by the progressive method by heating in steps of 100°C up to 700°C in an electric furnace

monitored by a thermocouple. The heating and subsequent cooling to room temperature was performed in field-free space provided by a three-axis 2-m Helmholtz coil system monitored by a Schonstedt Hecon-Helmholtz Control System, periodically checked with a Forster Millioerstedmeter. In general, thermal demagnetization was less satisfactory than alternating field demagnetization (with a few minor exceptions). In either case, demagnetization was carried out in steps of 100 Oe or 100°C up to peaks of 700 Oe or 700°C, respectively. Stability in the intensity and in the direction of magnetic moments was reached by either 500 Oe or 500°C in most cases. Treatment in 500 Oe peak field generally marked an inflection point in the J/J_{NRM} curves beyond which the intensity did not decrease as rapidly as it did prior to 500 Oe, and the direction of the magnetic moment stabilized. (J is the residual total moment after treatment, J_{NRM} is the uncleaned magnetic moment).

3. Structural corrections

The importance of the structural corrections in the derivation of a geomagnetic pole from these igneous rocks cannot be overemphasized. The igneous rocks were intruded into a sedimentary basin to the southeast and were transported into their present position within the nappe as a result of movements associated with the Taconic orogeny [3,4]. In the course of this movement the rocks were folded and overturned [5]. Due to the limited number of outcrops, the structural corrections used here are the only ones available for the sampled units. Unfortunately, the best outcrops of the Martinsburg Formation in the quadrangle do not occur in the same area as the outcrops of diabase. Of the structural parameters, the absolute verification of overturned bedding at all sites was the parameter most seriously impaired by the lack of good outcrops. The bedding in the Cambro-Ordovician carbonates immediately to the south has been shown to be overturned [6] and the available evidence (baked contacts adjacent to basalts) points to the existence of overturned beds in the shale [2].

The structural parameters used were derived primarily from Schmidt net compilations of the field data (see Table 1). With the exception of the axial plane from site 2, the axial planes and fold axes had to be derived from the attitudes of bedding along strike. The remaining axial planes were approximated to be the plane bisecting the angle between the opposite limbs of the fold. The fold axis in each case was considered to be the point of the greatest concentration of fold traces (beta diagram). The structural rotations performed by the computer were checked on the stereo-net. The bed and axial plane were plotted as poles to the planes while the fold axis was plotted as a line, all in the lower hemisphere. The declination and inclination of the magnetic moment was plotted as a line in the appropriate hemisphere (positive inclination in the lower hemisphere and negative inclination in the upper hemisphere). All rotations were performed as outlined in chapter 3 of Turner and Weiss [7], with the exception of the declination-inclination point, which had to be treated separately. When the latter point is in the lower hemisphere, it is rotated in the same sense as the fabric elements. When the point is in the upper hemisphere, the apparent direction of rotation is opposite that of the fabric elements. In either case, when a rotated declination-inclination point reaches the primitive circle, the point reverses direction and completes the required rotation along the same small circle on which the rotation began. The complete unfolding of a fold using the structural data from site 1 and the NRM declination and inclination of sample 32, is illustrated in Fig. 1. The contact between the diabase and shale was added to the rotations and indicated that the exposure at site 1 is a discordant intrusion. An additional structural correction for the rotation of the nappe cannot be made with the structural data available, adding uncertainty to the calculation of the pole position.

TABLE 1

Structural corrections

Site	Azimuth/amount of dip of bed	Azimuth/ plunge of fold axis	Azimuth/amount of dip of axial plane
1	276/142	025/13	298/78
2	327/133	161/47	150/78
4	327/133	161/47	150/48
7	326/158	233/20	315/60

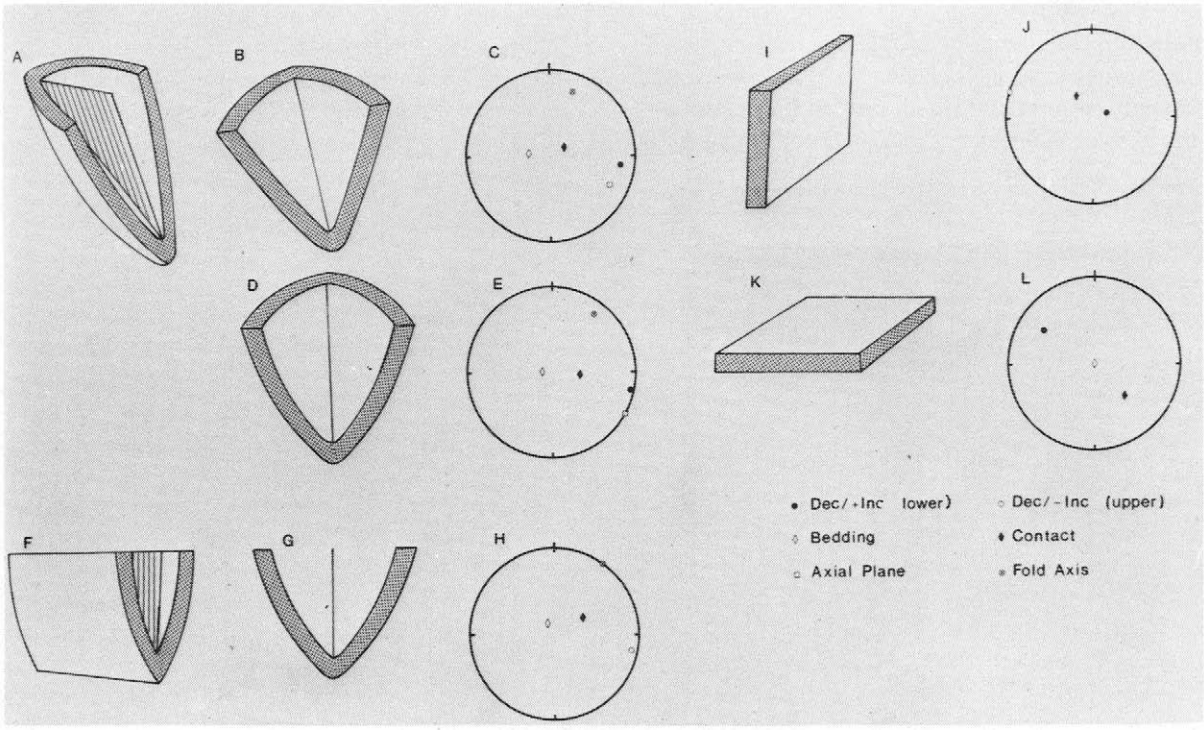


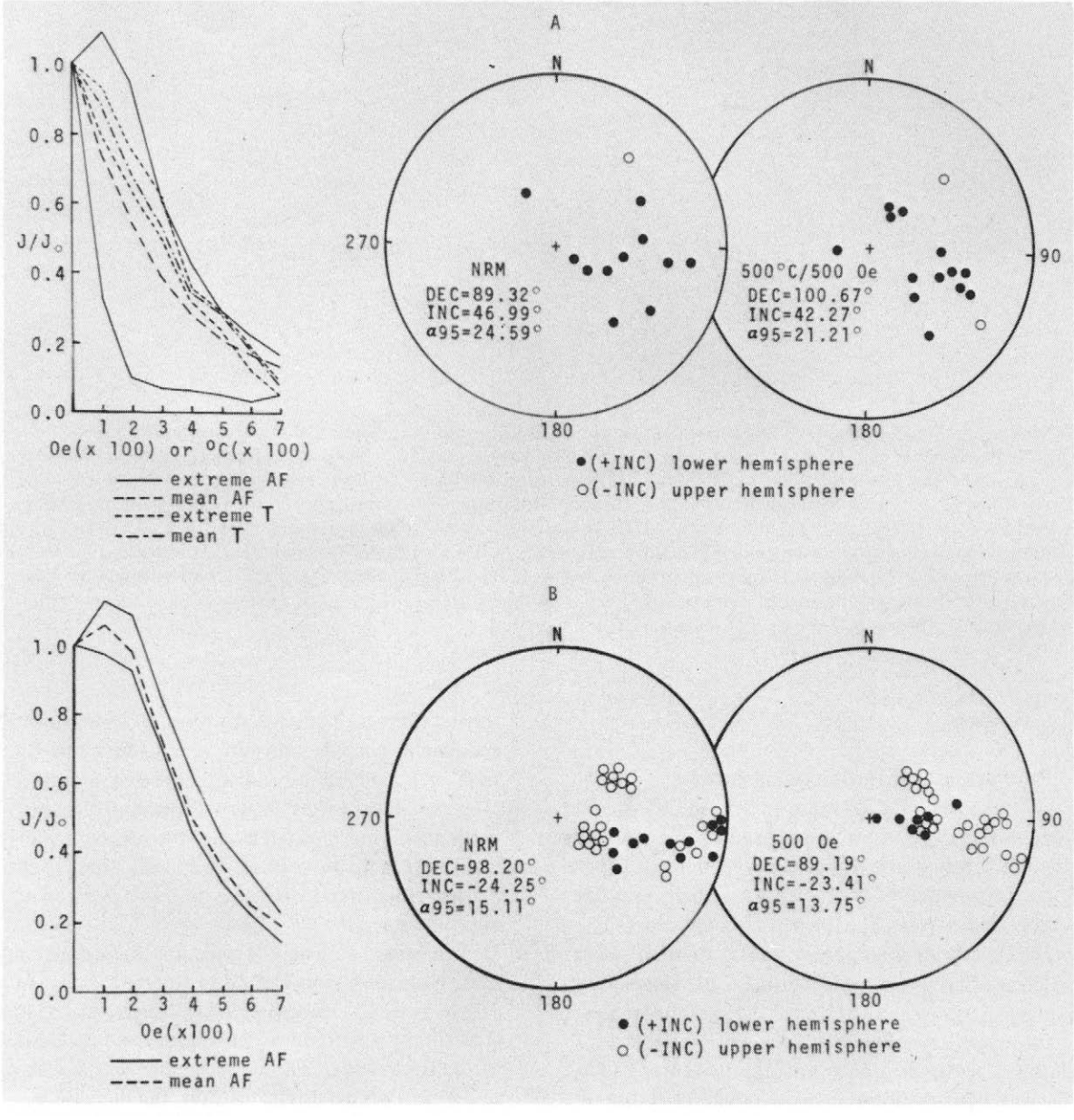
Fig. 1. The unfolding of the fold. All poles are in the lower hemisphere with the exception of the declination/inclination pole which can be in either hemisphere (open circle for upper hemisphere) as the fold axis and axial plane were corrected for they were removed from further rotations. A. Side view of the original fold. B. View parallel to the axial plane. C. Equal-area projection of the fabric elements and the direction of magnetization. D. Rotation of the axial plane to the vertical position. E. Projection of the fabric elements after rotation shown in D. F. Side view of the fold following the rotation of the fold axis to horizontal position. G. View parallel to axial plane for the same rotation as in F. H. Projection of the fabric elements following the rotation shown in F and G. I. Rotation of the overturned bedding to a vertical position. J. Projection of the fabric elements from I. K. Rotation of the bedding to its original position. L. Projection equivalent to K showing the original direction of the magnetization and the original attitude of the contact of the intrusion with the bedding.

4. Discussion

Prior to the final evaluation of the data and the structural corrections, it must be determined whether the rocks were able to retain a magnetic moment and whether the magnetic moment is a remanent of the original formation of the rocks. An evaluation of the Koenigsberger ratio, Q_N (using J_{NRM} and $H = 0.5$), showed that they were always greater than unity, indicating that pseudo-single-domain and single-domain titanomagnetites were the probable carriers of the magnetic moments [8]. Upper Paleozoic remagnetization may be a problem for Lower Paleozoic rocks [9]. McElhinny and Opdyke [10] dismissed Creer's [9] theory for most rocks except red sediments. There is, however, a possibility that the rocks studied here were remagnetized as a result of later orogenic heating. Zen [11] believed that the metamorphism reflected in the igneous rocks was the product of a Taconic event. It is likely, then, that the Lower Ordovician rocks were not remagnetized at any time since the Ordovician.

The curves showing the decay of the residual magnetic moment, J, normalized to the NRM moment, J_{NRM}, as well as the plots of the distribution of the structurally uncorrected declinations and inclinations of the NRM and of selected cleaning steps, are shown in Fig. 2. Two persistent features, the increase in

intensity at 100 Oe and the previously discussed inflection in the slope at 500 Oe, appear on some of the demagnetization graphs. In some cases where the intensity increased during alternating field cleaning an opposing moment was removed by cleaning in 100 Oe. It would be convenient to assign this magnetization to an IRM or VRM acquired much later than the primary magnetization. However, the persistence of the peak between samples, and the fact that its removal did not significantly shift the direction of the magnetic moment, seem to indicate that the weak, erased component may have been closely related to



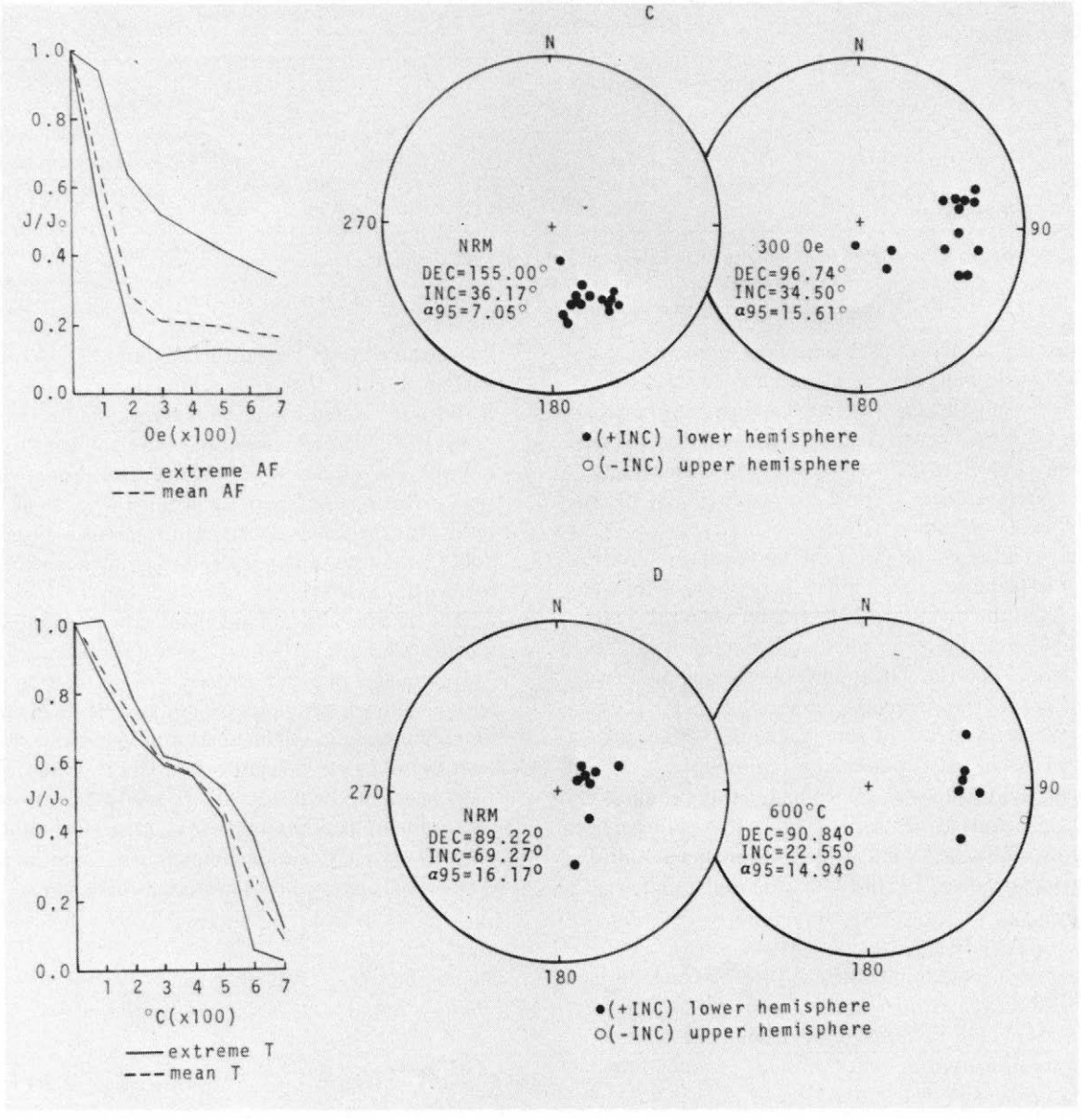


Fig. 2. The characteristic demagnetization curves and the distribution of the directions of magnetization of the NRM and at the indicated cleaning step (neither the NRM nor the demagnetization step have been corrected for structure in this example). A. Site 1. B. Site 2. C. Site 4. D. Site 7.

the primary magnetization. Perhaps this component was a low-temperature partial TRM locked in during a reversal of the earth's magnetic field while the diabase was cooling. Conclusive proof of the lack of severe alteration of the magnetic minerals cannot be established, but the similarity of the characteristic magnetization between sites and samples seems to indicate that the magnetization is primary.

It must be stressed that without the structural corrections the results presented here would be

TABLE 2

Paleomagnetic data

Treatment	Samples (specimens)	Latitude (°N)	Longitude (°E)	D (°)	I (°)	k	α_{95} (°)	dp (°)	dm (°)
Uncorr. NRM	26 (133)	71.4	141.9	105.3	37.4	1.46	40.9	62.8	71.7
Corr. NRM	26 (133)	48.0	141.1	308.2	−1.4	2.44	23.4	20.0	30.6
Corr.-cleaned	30 (133) *	35.8	159.8	318.3	−1.0	4.70	13.6	9.1	15.8

* Four samples were divided into two groups and cleaned by thermal and alternating field methods.

meaningless. With this in mind, the improvement in the results using the corrections may be evaluated. The mean poles and associated statistical parameters for both the uncorrected and corrected sample NRM poles (as well as the best estimate of the cleaned pole) are given in Table 2. While none of the NRM statistics indicate even a moderate degree of agreement between the samples (not totally unexpected for uncleaned results), note that there was an improvement in the precision of the results when the structural corrections were applied. From this it can be assumed that the structural corrections approached the goal of improving the between site correlation of results. The lack of data concerning within-site correlations and differential rotation between the shale and the diabase are reasons that the resolution of the direction was not as great as it could have been. (The shale at site 2 did appear to have been folded against the dike, but the data were not complete enough to allow correction.) The internal structural modifications of the nappe during transport cannot be fully accounted for at the present time. In summary, the structural corrections did significantly improve the resolution of the data. The ultimate justification for the corrections will come when more results for the Lower Ordovician are available (see Table 3 for a summary of Cambro-Ordovician poles for North America).

The declinations and inclinations for the individual samples are given in Table 4. Since the site was close to the equator in the Ordovician, the delineation of normal and reversed poles is questionable. Some of the discrepancy in the inclination could be due to a self-reversal in the magnetization. The samples from site 7 have a declination which is nearly 180° from the declinations of the other sites, suggesting either a magnetic reversal or an incomplete structural correction. It was observed, however, that there was no

TABLE 3

Lower Paleozoic paleomagnetic poles for North America

Age	Latitude (°N)	Longitude (°E)	Location	Reference
Late Ordovician	35	126	New Jersey	[20]
Late Ordovician	35	–	Kentucky	[13]
Middle Ordovician	36	114	New York	[10]
Early Ordovician	36	160	Pennsylvania	this paper
Cambro-Ordovician	37	122	Colorado	[19]
Late Cambrian	6	159	Texas	[18]
Late Cambrian	1	168	Missouri	[17]
Late Cambrian	4	164	Oklahoma	[16]
Early Cambrian	10	124	New Brunswick	[15]
Late Precambrian-Cambrian	12	156	Colorado	[14]

TABLE 4

Individual samples used in compilation of pole

Sample (specimens)	Site	Treatment	D (°)	I (°)	k	α_{95} (°)	dp (°)	dm (°)
1 (4)	1	500°C	350.4	−33.2	81.5	10.2	6.6	11.6
1 (1)	1	700 Oe	336.1	−39.5	–	–	–	–
2 (2)	1	600°C	331.7	−6.5	3.0	65.4	33.0	65.8
4 (9)	2	500 Oe	9.0	55.0	293.6	3.0	3.0	4.3
5 (7)	2	500 Oe	273.5	−19.5	22.5	13.0	7.1	13.6
9 (5)	1	200 Oe	324.0	−59.2	4.6	39.8	44.6	59.6
9 (6)	1	400°C	348.5	−0.8	2.5	53.5	33.3	59.7
10 (2)	1	700°C	179.1	−13.6	2.8	67.9	35.4	69.3
11 (9)	1	600°C	308.1	−19.4	12.5	15.2	8.3	15.8
14 (5)	7	600°C	121.9	−10.6	17.9	18.6	9.6	18.9
15 (1)	7	500°C	117.8	−6.9	–	–	–	–
19 (1)	7	500°C	92.5	−2.1	–	–	–	–
19 (1)	7	600 Oe	96.8	−17.7	–	–	–	–
21 (5)	4	700 Oe	304.0	20.0	61.1	9.9	5.4	10.3
23 (1)	4	100 Oe	238.5	−3.0	–	–	–	–
24 (7)	4	700 Oe	312.2	−3.3	6.34	26.0	13.0	26.0
24 (5)	1	100 Oe	299.6	−1.3	120.7	7.0	3.5	7.0
26 (13)	1	500 Oe	277.2	43.1	265.9	2.6	2.0	3.2
27 (2)	1	700 Oe	312.4	3.7	158.6	7.3	3.7	7.3
28 (3)	1	700 Oe	27.5	44.4	162.0	9.7	7.9	12.2
32 (1)	1	500 Oe	318.9	−3.7	–	–	–	–
35 (5)	2	600 Oe	323.4	38.4	312.5	4.3	3.1	5.2
36 (3)	2	600 Oe	16.3	30.0	494.6	5.6	3.4	6.2
37 (3)	2	600 Oe	329.2	41.9	43.8	18.9	14.2	23.1
40 (6)	1	500 Oe	309.7	11.7	14.0	18.6	9.6	18.9
41 (4)	1	600 Oe	312.6	4.1	647.5	3.6	1.8	3.6
42 (7)	1	500 Oe	273.5	2.4	113.2	5.7	2.9	5.7
43 (3)	1	300 Oe	284.8	−26.5	86.7	13.3	7.8	14.4
45 (5)	2	600 Oe	304.2	39.5	95.3	7.9	5.7	9.4

apparent difference in the petrology between samples with positive inclination and those with negative inclination. Samples with positive inclination yield a northern hemisphere paleomagnetic pole which is close to the northern hemisphere pole obtained from samples with negative inclinations, making field reversal during the period of intrusion of the rocks a distinct possibility. The average of the cleaned and corrected sample poles results in an Early Ordovician paleomagnetic pole at 159.8°E, 35.8°N (α_{95} = 13.6°) (see Table 2). The position of the pole in relation to the continuous apparent polar wander path for North America is shown in Fig. 3. The Cambrian and Ordovician poles are from Table 3 and the post-Ordovician poles are those compiled by Van Alstine and de Boer [12]. (The Middle to Late Ordovician results from Bartman et al. [13] are from a borehole and were not used in Fig. 3.) The position of the pole off a continuous path between the Cambrian and Middle Ordovician poles suggests that there may indeed have been uncorrected rotation in the horizontal plane. We cannot be certain if uncorrected rotation did occur until other Early Ordovician poles from North America are available. For the present, this study demonstrates that extreme deformation is not necessarily an obstacle in paleomagnetic studies, provided the regional structural history can be unraveled.

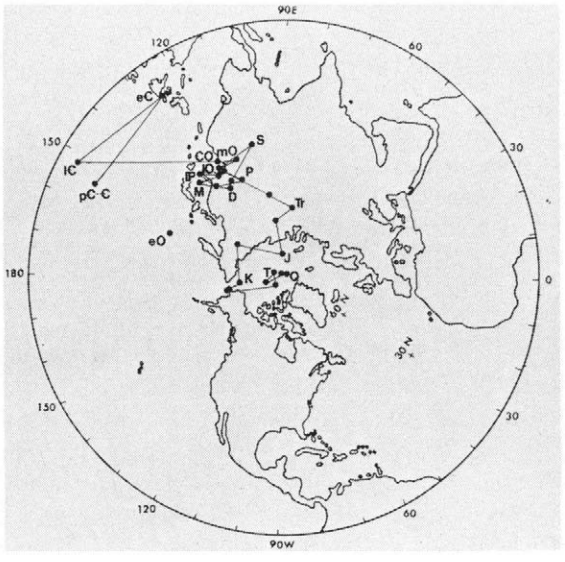


Fig. 3. Polar wandering curve for North America including the pole for the Ordovician reported in this paper (modified after Van Alstine and de Boer [12]).

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